

■ The Magic of Lenses ■

How They Bend Light and Make Things Bigger!

Subject: Science — Light & Optics	Grade: 4
Duration: 45–60 minutes	Topics: Refraction, Convex & Concave Lenses

■ Learning Objectives

By the end of this lesson, students will be able to:

1. Define a lens and explain what refraction means.
2. Describe the differences between convex and concave lenses.
3. Identify everyday objects that use convex or concave lenses.
4. Explain how each lens type affects the path of light rays.

■ Materials Needed

- A glass of water • A pencil or straw • Magnifying glass (if available)
- Reading glasses • Myopia glasses • Student notebooks & coloured pencils

SECTION 1 — What Is a Lens?

Key Concept

A **lens** is a clear, curved piece of glass or plastic that **bends light**. Because it is transparent (lets light pass through), light rays change direction as they pass through it, allowing us to see objects bigger, smaller, or more clearly.

■ Key Vocabulary

Word	Meaning	Example
Lens	A curved piece of glass/plastic that bends light	Glasses, microscope, camera
Transparent	Allows light to pass through it	Glass, clear water, plastic wrap
Refraction	The bending of light as it moves from one material to another	Straw appearing broken in water
Convex	Curved outward; thicker in the middle	Magnifying glass, reading glasses
Concave	Curved inward; thinner in the middle	Myopia glasses, door peephole
Converge	Light rays coming together at a single point	How convex lenses focus light
Diverge	Light rays spreading apart	How concave lenses spread light

SECTION 2 — Refraction: Light Plays Tricks!

Light normally travels in a straight line. But when it passes from one material (like air) into another (like water or glass), it **changes speed** — and that causes it to bend. This bending is called **refraction**.

■ Classroom Demonstration: Lemon Water Trick

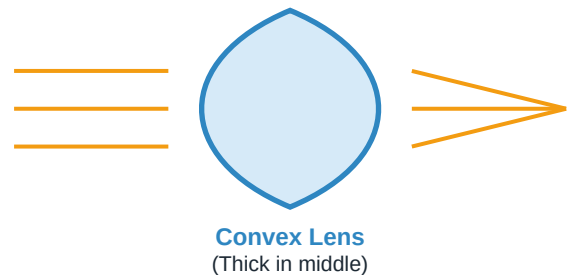
Place a pencil or straw in a glass of water and look from the side. The pencil appears *bent or broken* at the waterline — but it is not! Light bends (refracts) when it moves from water into air, creating an optical illusion. This same principle is what makes lenses work.

What to observe: The apparent break, its position at the water surface, and how moving your viewing angle changes what you see.

SECTION 3 — The Convex Lens (The Magnifier)

What does it look like?

A convex lens is **thicker in the middle** and thinner at the edges — like a *laddoo* or a football seen from the side.



How does it work?

When parallel light rays pass through a convex lens, they are bent **inward (converge)** and meet at a point called the **focal point**. This is why objects look bigger through a convex lens.

■ Memory Trick

"Convex Comes OUT — Converges — = BIG!"

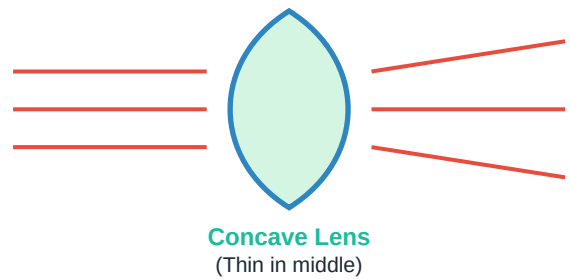
■ Real-Life Uses of Convex Lenses

Object	How Convex Lens Helps
Reading Glasses (Dadi's chashmey)	Magnifies text so near-sighted elders can read comfortably
Magnifying Glass	Concentrates light to make tiny objects appear much larger
School Microscope	Uses multiple convex lenses to magnify tiny specimens up to 1000×
Mobile Camera Lens	Focuses light onto the image sensor to create sharp photographs
Human Eye Lens	A natural convex lens that focuses light onto the retina
Projector	Enlarges and projects images onto a screen

SECTION 4 — The Concave Lens (The Shrinker)

What does it look like?

A concave lens is **thinner in the middle** and thicker at the edges — like a cooking bowl (*karahi*) or a cave that curves inward.



How does it work?

When light rays pass through a concave lens, they are bent **outward (diverge)** — spreading away from each other. This makes objects appear **smaller** and further away.

■ Memory Trick

"**Concave CAVES IN** like a bowl — = **SMALLER!**"

■ Real-Life Uses of Concave Lenses

Object	How Concave Lens Helps
Myopia (Short-sight) Glasses	Spreads light outward so far-away objects appear sharp
Door Peephole (Magic Eye)	Gives a wide-angle view of who is outside the door
Wide-Angle Camera Lens	Captures a very broad field of view in photography
Flashlight / Torch	Spreads the light beam wider to illuminate a larger area
Telescope eyepiece	Used alongside convex lenses to correct image orientation

SECTION 5 — Convex vs Concave: Side-by-Side Comparison

Feature	Convex Lens ■	Concave Lens ■
Shape	Thicker in the middle	Thinner in the middle
Resembles	Laddoo / football side-view	Karahi / cave / bowl
Effect on rays	Converges (brings together)	Diverges (spreads apart)
Image size	Makes things BIGGER	Makes things SMALLER
Focal point	Real focal point (in front)	Virtual focal point (behind lens)
Common use	Magnifying glass, reading specs	Myopia specs, door peephole
Eye condition	Helps hyperopia (far-sighted)	Helps myopia (short-sighted)

SECTION 6 — ■■ Lens Detective Activity!

Look at each object below. Identify whether it uses a **Convex lens**, **Concave lens**, or **No lens**. Write your answer in the blank. Discuss with a partner before checking!

#	Object	Your Answer	Correct Answer
1	Dadi's reading glasses	_____	Convex
2	Student's myopia glasses	_____	Concave
3	Magnifying glass	_____	Convex
4	Window glass (flat)	_____	No lens (flat glass)
5	Door peephole / Magic Eye	_____	Concave
6	Mobile camera lens	_____	Convex
7	Bathroom flat mirror	_____	Not a lens (mirror)
8	School microscope	_____	Convex
9	Sunglasses (tinted flat)	_____	No lens
10	Torch / flashlight reflector	_____	Concave

SECTION 7 — Student Worksheet

Name: _____ Date: _____ Class: _____

Part A — Fill in the Blanks

1. A lens is a _____ piece of glass or plastic that bends _____.
2. The bending of light when it passes from one material to another is called _____.
3. A convex lens is _____ in the middle and makes things look _____.
4. A concave lens is _____ in the middle and makes things look _____.
5. When light rays come together at a point, we say they _____ (converge/diverge).
6. A _____ lens is used in reading glasses for elders (hyperopia).
7. A _____ lens is used in glasses for myopia (short-sightedness).

Part B — True or False

Statement	T / F	Answer
A concave lens makes objects appear larger.	_____	False
Refraction is the bending of light.	_____	True
A convex lens converges light rays.	_____	True
Window glass (flat) is a type of lens.	_____	False
A door peephole uses a convex lens.	_____	False — it uses a concave lens
The human eye lens is convex.	_____	True

Part C — Short Answer Questions

- ◆ 1. Explain in your own words what refraction is. Give one example.

Answer:

Suggested: Refraction is when light bends as it moves from one material to another (e.g., air to water). Example: a straw appears broken in a glass of water.

- ◆ 2. How is a convex lens different from a concave lens? (mention shape and effect on light)

Answer:

Suggested: Convex: thicker in middle, converges light, makes things bigger. Concave: thinner in middle, diverges light, makes things smaller.

◆ **3. Why do people with myopia (short-sightedness) need concave lenses?**

Answer:

Suggested: In myopia the eye focuses light in front of the retina. A concave lens diverges the light first so it focuses correctly on the retina.

◆ **4. Name two objects you use daily that contain a convex lens.**

Answer:

Suggested: Accept: magnifying glass, mobile camera, reading glasses, school microscope, eye.

SECTION 8 — Homework: Lens Hunt!

Explore your home and find **at least 3 objects** that contain a lens. For each object, draw a picture in your notebook and answer the questions below.

Object Name	Draw It Here	Convex or Concave?	What does it help you do?
1.			
2.			
3.			
BONUS 4.			

■ Homework Tips

- Ask a family member to help you look! Make it a fun scavenger hunt.
- Check the kitchen (torch, magnifying glass), bathroom, bedrooms.
- Remember: flat glass (windows, bathroom mirrors) is NOT a lens.
- Use coloured pencils for your drawings — be creative and have fun!

SECTION 9 — Lesson Recap & Vocabulary Flashcards

What Did We Learn Today?

■ Refraction	Light bends when it passes from one material into another (e.g., air → water).
■ Convex Lens	Shaped like a laddoo. Thick in middle. Converges light. Makes things BIGGER.
■ Concave Lens	Shaped like a bowl/karahi. Thin in middle. Diverges light. Makes things SMALLER.
■ Transparent	A material that lets light pass through it (glass, water, clear plastic).
■ Real Life	Lenses are everywhere — glasses, cameras, microscopes, telescopes, our own eyes!

✂️ ■ Cut-Out Vocabulary Flashcards

Cut these out and keep them in your science folder. Quiz yourself or a friend!

LENS A clear curved piece of glass or plastic that bends light to make things appear bigger or smaller.	REFRACTION The bending of light when it passes from one material into another (e.g., air into water).
CONVEX A lens that is thicker in the middle. It converges (brings together) light. Makes objects look BIGGER.	CONCAVE A lens that is thinner in the middle. It diverges (spreads) light. Makes objects look SMALLER.
TRANSPARENT A material that allows light to pass through it clearly. Example: glass, water, clear plastic.	CONVERGE When light rays bend inward and come together at a single point (focal point).

TEACHER'S NOTES & ASSESSMENT GUIDE

Lesson Flow Suggestions (45-min class)

Time	Activity	Notes
0–5 min	Hook: Show a magnifying glass. Ask: 'Why do things look bigger?'	Elicit prior knowledge. Let students try the glass.
5–10 min	Introduce: What is a lens? Key vocabulary display.	Use the vocab table. Ask students to repeat each word.
10–15 min	Demo: Straw/pencil in water — refraction explained.	Have students predict first. Discuss what happened.
15–25 min	Teach convex vs concave with diagrams and memory tricks.	Use real glasses if available. Students sketch lens shapes.
25–35 min	Lens Detective Activity — identify lens types in objects.	Pair work first, then class discussion.
35–42 min	Worksheet Part A & B (fill-in-the-blanks + T/F).	Walk around to support; don't reveal answers yet.
42–45 min	Recap and introduce homework (Lens Hunt).	Give students 2 minutes to discuss what they found interesting.

Assessment Rubric

Criterion	Excellent (4)	Good (3)	Developing (2)	Needs Support (1)
Vocabulary	Uses all 5 key words correctly	Uses 3–4 words correctly	Uses 1–2 words	Cannot use key words
Lens identification	Correctly identifies all 8 objects	Correctly identifies 6–7	Correctly identifies 4–5	Identifies fewer than 4
Explanation quality	Clear, detailed, own words	Mostly clear, minor errors	Partial explanation	Confused or off-topic
Homework (Lens Hunt)	4+ objects, detailed drawings	3 objects, good drawings	2 objects, basic drawings	1 or no objects

■ Differentiation Tips

Extension: Research how telescopes or the human eye use multiple lenses together.

Support: Provide a word bank for the fill-in-the-blank section.

Visual learners: Use real objects — bring in actual glasses, magnifying glass.

Cultural connect: Use familiar Indian examples (laddoo, karahi, chashmey).

Home connect: Lens Hunt homework engages families in science exploration.